Outage Investigation Report — OIR-2025-006

Field	Value
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Report Date	2025-05-19
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Region	NW-PowerPool
Site	NW-PowerPool-WIND-014
Asset	WIN-10058 (GE Renewable Energy, wind_turbine)
Outage ID	OUT-2025-0433
Classification	Forced outage — soc_drift
Distribution	NW-PowerPool Operations, Reliability Engineering, OEM Liaison, Vendor Management

1. Outage Summary

WIN-10058 declared a forced outage on 2025-05-02 at 15:34 PDT after the unit's grid-interface battery buffer drifted out of spec and the protection scheme isolated the turbine. The unit returned to service on 2025-05-12 at 02:34 PDT, total duration **227 hours** — the longest single-unit outage at NW-PowerPool-WIND-014 in FY2025. Lost generation: **2,506.8 MWh**.

This is the **opening event of the Spring Vibration Cascade** (refer to RCA-2025-001 for the broader analysis). Within 4 weeks NW-PowerPool would log 19 forced outages across WIND-014 and WIND-024, totaling 11,360 MWh of lost generation. The cascade is traceable to the Q1 2025 budget freeze and to deferred preventive work on the GE fleet at WIND-014.

2. Timeline of Events

Timestamp (PDT)	Event
2025-05-01 22:10	Vibration RMS on WIN-10058 main shaft trending upward — 4.9 mm/s, baseline 3.6. Standard MS-2025-03 alert threshold 6.5.
2025-05-02 11:20	SoC drift on buffer Bank A-1 widens from 0.6% to 2.4%. CSM-7 §4.2 classifies "advisory" up to 2.0%.
2025-05-02 15:34	Protection trip on under-voltage co-incident with vibration RMS at 6.7 mm/s. Forced outage declared.
2025-05-02 17:55	CPS dispatched from Bend, OR — concurrent NW-PowerPool-WIND-024 work limits crew availability. ETA 9 hours. EPA-2024-11 invoked at 18:45.
2025-05-02 22:10	Apex Emergency Grid Solutions accepts dispatch. Onsite at 02:00 the following morning.
2025-05-03 — 2025-05-09	Bearing inspection — outer race spalling on main bearing. BMS firmware on Bank A-1 confirmed running 2.7.1 (recall: same defect identified in OIR-2025-001 for SUB-10629). Apex recommends bearing replacement and BMS firmware upgrade.
2025-05-09 — 2025-05-12	Bearing replacement, BMS upgrade to 2.7.4, vibration baseline retest.
2025-05-12 02:34	Returned to service.

3. Telemetry Observations

- Vibration RMS at trip: 6.7 mm/s (Cascadia MS-2025-03 limit 6.5; GE OEM Manual GRE-2200 threshold 8.0).
- SoC drift on Bank A-1 at trip: 2.4% (CSM-7 advisory > 2.0).
- Outer-race bearing inspection: spalling pattern consistent with 4–6 months of operation past the MS-2025-03 reinspection interval.
- Bank A-1 BMS firmware: 2.7.1 (defect noted in Hitachi Service Bulletin SB-HE-2024-09; same issue observed on SUB-10629 in OIR-2025-001).
- GE OEM Manual GRE-2200 specifies main-bearing reinspection at 8,000 operating hours. Cascadia MS-2025-03 reduced this to 6,000 hours after the 2023 firmware refresh changed load profile. The unit was at 7,420 hours since last inspection. See CAR-2025-003 (GE manual obsolescence).

- Ambient: 14 °C, SW wind 9–13 m/s. Conditions within norm.

The April 2025 quarterly main-bearing inspection (WO-2025-04022) was on the Q1 deferred list and not yet rescheduled at the time of the event.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Onsite diagnostics, main bearing replacement, BMS firmware upgrade, vibration retest	$441,000
GE OEM (technical advisory remote)	Procedure review, warranty acknowledgement	$22,500
Total		**$463,500**

CPS scheduled equivalent estimate (issued 2025-05-15 retrospective benchmark): $112,800. The 4× ratio is consistent with the broader cost-arbitrage pattern; see EMR-2025-003.

5. Preliminary Root Cause

Direct cause: outer-race bearing spalling from operation past the MS-2025-03 reinspection interval, combined with BMS firmware defect that allowed sustained SoC drift to propagate to the protection scheme.

Contributing causes:
1. WO-2025-04022 (main-bearing reinspection) was deferred under the Q1 budget freeze.
2. Documentation conflict: GE OEM Manual GRE-2200 inspection interval (8,000 h) is invalidated by the 2023 firmware load profile change. CAR-2025-003 has been open since the firmware refresh; not yet implemented at the asset level.
3. Bank A-1 BMS firmware 2.7.1 with known cell-balancing defect — the same firmware version implicated in OIR-2025-001 for SUB-10629 four weeks earlier. The fleet-wide BMS upgrade plan has been open under CAR-2024-118 since December 2024.

6. Recommendations

1. Mobilize emergency fleet-wide audit of main-bearing inspection status across NW-PowerPool-WIND-014. Track under CAR-2025-002 (Q1 freeze recovery).
2. Upgrade BMS firmware to 2.7.4 across all NW-PowerPool battery buffers within 30 days. Track under CAR-2024-118.

3. Reconcile GE OEM Manual GRE-2200 inspection intervals with the 2023 firmware load profile. Refer to CAR-2025-003. Internal engineering position: 6,000 h interval is correct; GE is invited to issue an updated manual.
4. Add WIN-10058 to the high-watch list. This unit's failure profile is the leading indicator of the Spring Vibration Cascade — see RCA-2025-001.
5. Cross-reference OIR-2025-007 (TRA-10561, same period, vibration_excess) for the second cascade event.

Signatures

Role	Name	Date
Author	Davis Ngo, Reliability Engineer I	2025-05-19
Plant Manager	Priya Ramachandran	2025-05-20
OEM Liaison	J. Bertelsen	2025-05-21

Distribution: NW-PowerPool Operations · Reliability Engineering · OEM Liaison · Vendor Management · CFO Office

References: OIR-2025-001 · OIR-2025-007 · CAR-2024-118 · CAR-2025-002 · CAR-2025-003 · EMR-2025-003 · RCA-2025-001 · GE OEM Manual GRE-2200 · Standard MS-2025-03 · CSM-7 · Hitachi Service Bulletin SB-HE-2024-09